CLASS 8 · JEE MAINS · TOP-10 QUESTION IMPACT REPORT

Pragya G8 (EXI) Achievers

Part Test-1 · which questions cost the top 10 the most

Exam date 08 Oct 2026
Campus HYD/Nallakunta DD Colony
Students analysed top 10 by program rank

In total, the top 10 lost **1,262 marks** – **340** from Maths, **617** from Physics and **305** from Chemistry.

Maths 27% | Physics 49% | Chemistry 24%

Together they could have scored 2,960 (296 each) and scored 1,698 – the 1,262 marks lost is the gap. Marks lost = marks not earned on wrong or skipped questions plus negative marks.

29 of 74
high-impact questions – lost by 5+ of the top 10

49%
of all lost marks came from Physics

Q67 10/10
lost by every student

170 / 300
top-10 average score (range 147–204)

1 Where the damage happened

Marks the top 10 gave away in each subject, split into marks lost on **wrong answers** (the question's marks plus the negative mark) and marks left on **skipped** questions.

SUBJECT	SPLIT ■ wrong ■ skipped	MARKS LOST	SHARE	ACCURACY*	HIGH-IMPACT
Maths	220 on 44 wrong answers (incl. 44 negative) · 120 on 30 skipped	**340** ≈ 34 per student	27%	79%	**7** of 24 Qs
Physics	385 on 77 wrong answers (incl. 77 negative) · 232 on 58 skipped	**617** ≈ 62 per student	49%	60%	**16** of 25 Qs
Chemistry	245 on 49 wrong answers (incl. 49 negative) · 60 on 15 skipped	**305** ≈ 30 per student	24%	79%	**6** of 25 Qs

* Accuracy = right answers ÷ questions attempted.

BY QUESTION TYPE	MATHS	PHYSICS	CHEMISTRY
Single correct marking +4 / −1	Q1–20 **19%** wrong · **12%** skipped	Q26–45 **34%** wrong · **18%** skipped	Q51–70 **21%** wrong · **4%** skipped
Numerical marking +4 / −1	Q21–25 **16%** wrong · **16%** skipped	Q46–50 **18%** wrong · **46%** skipped	Q71–75 **14%** wrong · **12%** skipped

Key takeaways

1. **Physics** did the most damage: **49%** of all marks lost by the top 10 (617 of 1262). 16 of its 25 questions were lost by 5 or more students.
2. **Q67** was lost by all 10 students, and **6 more** by 9 of 10 (Q16, Q35, Q36, Q43, Q47, Q49).
3. Biggest chapter-level loss: **Physics · Laws of Motion** – 243 marks across 8 questions (7 high-impact).
4. Skipping is concentrated in **Physics · Numerical**: 23 of 50 answers (46%) were left blank.
5. Negative marking alone cost the top 10 **170 marks** (about 17 per student).
6. Strong ground: all 10 answered **10 questions** correctly (Q2, Q8, Q10, Q15, Q34, Q52, Q63, Q64, Q68, Q71).

Answered wrong by 5+ students

Students attempted these and still got them wrong – points to a misconception or a trap worth re-teaching.

Q67 10 wrong | **Q16** 9 wrong | **Q36** 9 wrong | **Q41** 7 wrong
Q43 7 wrong | **Q5** 6 wrong | **Q45** 6 wrong | **Q3** 5 wrong
Q4 5 wrong | **Q30** 5 wrong | **Q40** 5 wrong | **Q65** 5 wrong

Left blank by 5+ students

Most students did not attempt these – points to an unfamiliar topic, low confidence, or running out of time.

Q35 8 skipped | **Q38** 8 skipped | **Q47** 8 skipped | **Q49** 7 skipped
Q14 5 skipped | **Q27** 5 skipped | **Q48** 5 skipped

2 Question-by-question impact

How many of the top 10 lost each question. Questions at or above the dashed line (lost by 5+) are **high-impact** and shown in full colour with their count.

Wrong answer · Skipped · ---- High-impact line (5 of 10) · Below the line (faded) · "del" = deleted / bonus question



3 Chapter hot-spots

Marks lost by the top 10, chapter by chapter. Subjects are ordered from most marks lost to least; within each subject, re-teach from the top.

Marks lost adds up what all 10 students gave away on a chapter's questions: **5 marks** for each wrong answer (4 not earned + 1 negative) and **4 marks** for each skipped one. Example: **Physics · Laws of Motion** has 8 questions, and the top 10 lost **243** marks on them – about **24 per student**. **Students per Q** = on average, how many of the 10 lost each question in that chapter (here 6.4 of 10).

CHAPTER	QS	HIGH-IMPACT QUESTIONS	MARKS LOST BY TOP 10	PER STUDENT	STUDENTS PER Q
1. Physics 617 marks lost · 49% of all lost marks · 16 high-impact questions					
Laws of Motion	8	Q30, Q40, Q41, Q42, Q43, Q44, Q45	243	≈ 24 marks	6.4 of 10
Kinematics	10	Q35, Q36, Q38, Q48, Q49	231	≈ 23 marks	5.2 of 10
Vectors	7	Q27, Q28, Q29, Q47	143	≈ 14 marks	4.6 of 10
2. Maths 340 marks lost · 27% of all lost marks · 7 high-impact questions					
Quadratic Equations	5	Q5, Q16	114	≈ 11 marks	4.8 of 10
Surds & Indices	11	Q17, Q23	101	≈ 10 marks	2.1 of 10
Sequences & Series	5	Q3, Q4	82	≈ 8 marks	3.4 of 10
Logarithms & Exponents	3	Q14	43	≈ 4 marks	3.3 of 10
3. Chemistry 305 marks lost · 24% of all lost marks · 6 high-impact questions					
Chemical Bonding	6	Q59, Q67, Q75	121	≈ 12 marks	4.2 of 10
Mole Concept & Basics	9	Q53, Q55	87	≈ 9 marks	2.1 of 10
Atomic Structure	5	None reached 5 · most missed Q57, Q58 (3/10)	49	≈ 5 marks	2.0 of 10
Periodic Classification	5	Q65	48	≈ 5 marks	2.0 of 10

4 High-impact questions at a glance

Every question lost by 5 or more of the top 10, one table per subject, worst first. The cropped question for each is in section 7.

Marks lost = marks the top 10 together gave away on that question: **5 marks** for each wrong answer (4 not earned + 1 negative) and **4 marks** for each skipped one. Example: **Q67** – 10 wrong × 5 = 50 marks. **Lost by** = how many of the 10 got it wrong or skipped it.

Critical lost by 9–10 **High** lost by 7–8 **Moderate** lost by 5–6

Maths

7 of 24 questions high-impact · **205** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q16	**Quadratic Equations** – Roots in GP, condition on discriminant	Single · +4 / −1	9/10	9	0	45
Q5	**Quadratic Equations** – Range of a rational quadratic expression	Single · +4 / −1	7/10	6	1	34
Q4	**Sequences & Series** – Identify AP / GP / HP from a surd expression	Single · +4 / −1	7/10	5	2	33
Q17	**Surds & Indices** – Cube-root surd identity (matrix match)	Single · +4 / −1	6/10	2	4	26
Q3	**Sequences & Series** – Product of GP terms; natural-number GP	Single · +4 / −1	5/10	5	0	25
Q23	**Surds & Indices** – Cube root of a surd expression	Numerical · +4 / −1	5/10	2	3	22
Q14	**Logarithms & Exponents** – Log-exponential equation; counting solutions	Single · +4 / −1	5/10	0	5	20

Physics

16 of 25 questions high-impact · **528** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q36	**Kinematics** – a(t) with modulus – area under a–t	Single · +4 / −1	9/10	9	0	45
Q43	**Laws of Motion** – Normal reaction across an inclined contact	Single · +4 / −1	9/10	7	2	43
Q49	**Kinematics** – Free fall then deceleration – find height	Numerical · +4 / −1	9/10	2	7	38
Q35	**Kinematics** – $a = kv^2$ – integrate to get v(s)	Single · +4 / −1	9/10	1	8	37
Q47	**Vectors** – Net displacement from multi-leg journey	Numerical · +4 / −1	9/10	1	8	37
Q41	**Laws of Motion** – Newton's 3rd law – fan blowing on own sail	Single · +4 / −1	8/10	7	1	39
Q45	**Laws of Motion** – Contact force between pushed blocks (statement type)	Single · +4 / −1	8/10	6	2	38
Q48	**Kinematics** – Distance from a–t graph	Numerical · +4 / −1	8/10	3	5	35
Q38	**Kinematics** – Resistance $\propto v^2$ – time inside plank	Single · +4 / −1	8/10	0	8	32
Q42	**Laws of Motion** – Extra tension when bird leaves the wire	Single · +4 / −1	7/10	4	3	32
Q27	**Vectors** – Vector components under axis rotation	Single · +4 / −1	7/10	2	5	30
Q30	**Laws of Motion** – Tension / normal reaction in two-block systems (matrix match)	Single · +4 / −1	5/10	5	0	25
Q40	**Laws of Motion** – Velocity over fixed distance vs mass ($v \propto 1/\sqrt{m}$)	Single · +4 / −1	5/10	5	0	25
Q28	**Vectors** – Resultant ⊥ smaller vector; find magnitudes	Single · +4 / −1	5/10	4	1	24
Q29	**Vectors** – Angles between three vectors summing to zero	Single · +4 / −1	5/10	4	1	24
Q44	**Laws of Motion** – Normal reactions on a cylinder resting on edges	Single · +4 / −1	5/10	4	1	24

Chemistry

6 of 25 questions high-impact · **182** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q67	**Chemical Bonding** – Strongest ionic bond (charge and size)	Single · +4 / −1	10/10	10	0	50
Q75	**Chemical Bonding** – Hybrid orbitals + lone pairs in NH_3	Numerical · +4 / −1	7/10	4	3	32
Q65	**Periodic Classification** – Largest radius among isoelectronic ions	Single · +4 / −1	6/10	5	1	29
Q55	**Mole Concept & Basics** – Laws of chemical combination	Single · +4 / −1	5/10	4	1	24
Q59	**Chemical Bonding** – VBT, orbital overlap, hybridisation statements	Single · +4 / −1	5/10	4	1	24
Q53	**Mole Concept & Basics** – Valence electrons in N^{3-} (mole → electrons)	Single · +4 / −1	5/10	3	2	23

5 Who lost which high-impact question

Each row is a student (campus order); each column a high-impact question. **Prog. rank** = rank in the program, **AIR** = All-India rank, **Total** = marks out of 300, **Avg** = average of the 6 weekly Mains tests (WTM), out of 300. ✗ = wrong, – = skipped, ✓ = correct.

✗ Wrong – Skipped ✓ Correct

STUDENT	Rank & score				Maths							Physics																Chemistry						Lost
	PROG. RANK	AIR	TOTAL	AVG /300	3	4	5	14	16	17	23	27	28	29	30	35	36	38	40	41	42	43	44	45	47	48	49	53	55	59	65	67	75	
Kolukula H. S. N. Hariharan	11	26	204	236.8	✗	✗	✓	✓	✗	–	✓	✓	✓	✗	✗	–	✗	–	✓	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗	15
M Charan Kumar Reddy	17	38	198	198.8	✓	✗	✓	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	–	✓	✓	✗	✓	✓	✗	–	–	✗	✓	✓	✓	✗	✗	–	11
Ruperao Shritan	26	59	188	228.8	✗	✓	–	–	✗	–	–	–	✓	✓	✓	–	✗	–	✓	✗	–	–	✗	–	–	✗	–	✓	–	✓	✓	✗	✗	21
Kavuru Sahithi	29	73	180	240.4	✓	✗	✗	–	✗	✓	✓	✗	✗	✓	✗	–	✗	–	✓	–	✗	✗	✓	✗	–	–	–	✗	✗	✗	✓	✗	✓	21
Fareeha Shaik	40	112	168	204.1*	✓	✗	✓	✓	✗	✗	–	–	✗	✓	✓	–	✗	✓	✗	✗	✓	✗	✓	✗	–	✗	✗	–	✗	✓	✗	✗	✓	19
M Jagadeesear Reddy	45	129	162	199.2	✗	–	✗	–	✗	–	✓	–	–	✗	✓	–	✗	–	✗	✗	✗	✗	✓	✓	–	✗	–	✓	✗	–	✗	–	–	24
Nakka Mahidhar	51	154	155	202.2	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓	–	–	✗	–	–	–	–	✓	✓	✗	✗	✗	23
Karthikeya Silver	55	178	148	210.3	✓	✓	✗	✓	✗	✓	–	–	✓	✓	✓	–	✗	–	✓	✗	–	✗	✗	✗	✗	✓	–	✗	✗	✗	✗	✗	✗	20
Aleti Dhanu Sai Sagar	56	180	148	174.0*	✓	–	✗	–	✗	–	✓	–	✓	–	✗	–	✗	–	✗	✗	–	✗	✗	–	–	–	–	✓	✓	✗	✓	✗	✓	22
Madipadiga Dheeraj	57	184	147	178.8	✗	✗	✗	✓	✗	✓	✗	✓	✗	✗	✗	–	✗	–	✗	✓	✓	✗	✗	✗	–	–	–	✗	✓	✗	✓	✗	–	22
Students who lost it					5	7	7	5	9	6	5	7	5	5	5	9	9	8	5	8	7	9	5	8	9	8	9	5	5	5	6	10	7	

6 Subject scores of the top 10

Scores rebuilt from each student's right / wrong / skipped answers (they match the official totals exactly). The weakest subject (lowest % of its maximum) is highlighted. **Subject maximum** leaves out the deleted / ungraded questions: Maths 96 = 100 minus Q20 (4).

STUDENT	PROG. RANK	AIR	TOTAL	AVG /300	Maths score · right / wrong / skipped	Physics score · right / wrong / skipped	Chemistry score · right / wrong / skipped	WEAKEST
Kolukula H. S. N. Hariharan	11	26	204	236.8	77/96 — 20 / 3 / 1	47/100 — 14 / 9 / 2	80/100 — 21 / 4 / 0	Physics
M Charan Kumar Reddy	17	38	198	198.8	74/96 — 19 / 2 / 3	60/100 — 16 / 4 / 5	64/100 — 17 / 4 / 4	Physics
Ruperao Shritan	26	59	188	228.8	54/96 — 14 / 2 / 8	48/100 — 13 / 4 / 8	86/100 — 22 / 2 / 1	Physics
Kavuru Sahithi	29	73	180	240.4	68/96 — 18 / 4 / 2	37/100 — 11 / 7 / 7	75/100 — 20 / 5 / 0	Physics
Fareeha Shaik	40	112	168	204.1*	63/96 — 17 / 5 / 2	38/100 — 12 / 10 / 3	67/100 — 18 / 5 / 2	Physics
M Jagadeesear Reddy	45	129	162	199.2	61/96 — 16 / 3 / 5	32/100 — 10 / 8 / 7	69/100 — 18 / 3 / 4	Physics
Nakka Mahidhar	51	154	155	202.2	57/96 — 16 / 7 / 1	35/100 — 11 / 9 / 5	63/100 — 17 / 5 / 3	Physics
Karthikeya Silver	55	178	148	210.3	67/96 — 18 / 5 / 1	36/100 — 11 / 8 / 6	45/100 — 14 / 11 / 0	Physics
Aleti Dhanu Sai Sagar	56	180	148	174.0*	48/96 — 13 / 4 / 7	20/100 — 7 / 8 / 10	80/100 — 21 / 4 / 0	Physics
Madipadiga Dheeraj	57	184	147	178.8	51/96 — 15 / 9 / 0	30/100 — 10 / 10 / 5	66/100 — 18 / 6 / 1	Physics

Average. "Avg" = the student's average over the 6 weekly Mains tests (WTM) before this Part Test, out of 300, from the campus Mains & Advance averages sheet. * Sat fewer weekly tests (tests sat / conducted): Fareeha Shaik (5/6), Aleti Dhanu Sai Sagar (3/6).
Ranks. Program rank = rank within Pragya G8 (EXI) Achievers across all campuses (from the final marks sheet; the top 10 are picked by this rank). **AIR** = All-India rank across the whole Class 8 programme (all batches, all campuses), from the institute's All India Marks Analysis sheet.
How to read this report. "Lost" means the student did not get the question right – either answered it wrong or left it blank. A question is **high-impact** when 5 or more of the top 10 lost it. **"Marks lost"** = marks the top 10 together gave away: **5 marks** for each wrong answer (4 not earned + 1 negative) and **4 marks** for each skipped one. "Per student" divides that by 10.
Final key applied. Deleted in the final key: Q20. Answer revised in the final key and counted as marked in the final marks sheet: Q49 (key 298), Q67 (key D). Left-out questions are marked "del" or "n/a" in the charts. Student totals and program ranks are as printed in the final marks sheet.
Chapter tags were assigned by reading each question; please correct any that your department classifies differently.

7 Maths – high-impact questions

7 questions, worst first
cropped from the question paper

Q16 Quadratic Equations – Roots in GP, condition on discriminant

Maths · Single correct · marking +4 / −1 · marks lost by the top 10: 9 wrong × 5 = 45

Lost by 9/10
9 wrong · 0 skipped

16. Let α, β be the roots of the equation $ax^2 + bx + c = 0$ and $\Delta = b^2 - 4ac$. If $\alpha + \beta, \alpha^2 + \beta^2, \alpha^3 + \beta^3$ are in G.P., then

(A) $\Delta = 0$ (B) $\Delta \neq 0$ (C) $b\Delta = 0$ (D) $c\Delta = 0$

Wrong: Kolukula Hema Shiva Naga Hariharan, Ruperao Shritan, Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Nakka Mahidhar, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Skipped: –

Q5 Quadratic Equations – Range of a rational quadratic expression

Maths · Single correct · marking +4 / −1 · marks lost by the top 10: 6 wrong × 5 + 1 skipped × 4 = 34

Lost by 7/10
6 wrong · 1 skipped

5. If x is real, then $\dfrac{x^2 + 2x + c}{x^2 + 4x + 3c}$ can take all real values if

(A) $0 < c < 2$ (B) $0 \le c \le 1$
(C) $-1 < c < 1$ (D) $0 < c < 1$

Wrong: Kavuru Sahithi, M Jagadeesear Reddy, Nakka Mahidhar, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Skipped: Ruperao Shritan

Q4 Sequences & Series – Identify AP / GP / HP from a surd expression

Maths · Single correct · marking +4 / −1 · marks lost by the top 10: 5 wrong × 5 + 2 skipped × 4 = 33

Lost by 7/10
5 wrong · 2 skipped

4. If a, b, c are in A.P., then $\dfrac{1}{\sqrt{a}+\sqrt{b}}, \dfrac{1}{\sqrt{a}+\sqrt{c}}, \dfrac{1}{\sqrt{b}+\sqrt{c}}$ are in

(A) A.P. (B) G.P. (C) H.P. (D) Neither A.P nor G.P nor H.P

Wrong: Kolukula Hema Shiva Naga Hariharan, M Charan Kumar Reddy, Kavuru Sahithi, Fareeha Shaik, Madipadiga Dheeraj

Skipped: M Jagadeesear Reddy, Aleti Dhanu Sai Sagar

Q17 Surds & Indices – Cube-root surd identity (matrix match)

Maths · Single correct · marking +4 / −1 · marks lost by the top 10: 2 wrong × 5 + 4 skipped × 4 = 26

Lost by 6/10
2 wrong · 4 skipped

17. Let $x = \left(a + \sqrt{a^2 + b^3}\right)^{\frac{1}{3}} + \left(a - \sqrt{a^2 + b^3}\right)^{\frac{1}{3}}$ and $x^3 + Abx - B.a = 0$, Then match

COLUMN-I with respect to COLUMN-II

	COLUMN-I		COLUMN-II
I)	A + B	P)	8
II)	A – B	Q)	9
III)	A^B	R)	1
IV)	B^A	S)	5
		T)	3

(A) I – S; II – R; III – Q; IV – P (B) I – S; II – T; III – P; IV – Q
(C) I – Q; II – P; III – T; IV – S (D) I – Q; II – P; III – S; IV – R

Wrong: Fareeha Shaik, Nakka Mahidhar

Skipped: Kolukula Hema Shiva Naga Hariharan, Ruperao Shritan, M Jagadeesear Reddy, Aleti Dhanu Sai Sagar

Q3 Sequences & Series – Product of GP terms; natural-number GP

Maths · Single correct · marking +4 / −1 · marks lost by the top 10: 5 wrong × 5 = 25

Lost by 5/10 — 5 wrong · 0 skipped

3. If $x_1, x_2, x_3, x_4, x_5, \ldots$ is a geometric progression of natural numbers and $x_1x_2x_3x_4 = 64$, then complete set of solutions of x_4 is

(A){1} (B){8} (C){1, 8} (D)cannot be determined

Wrong: Kolukula Hema Shiva Naga Hariharan, Ruperao Shritan, M Jagadeesear Reddy, Nakka Mahidhar, Madipadiga Dheeraj

Skipped: –

Q23 Surds & Indices – Cube root of a surd expression

Maths · Numerical · marking +4 / −1 · marks lost by the top 10: 2 wrong × 5 + 3 skipped × 4 = 22

Lost by 5/10 — 2 wrong · 3 skipped

23. If $\sqrt[3]{38\sqrt{14} - 100\sqrt{2}} = \sqrt{x} - \sqrt{y}$, then the value of $x - y$ is__?

Wrong: Nakka Mahidhar, Madipadiga Dheeraj

Skipped: Ruperao Shritan, Fareeha Shaik, Karthikeya Silver

Q14 Logarithms & Exponents – Log-exponential equation; counting solutions

Maths · Single correct · marking +4 / −1 · marks lost by the top 10: 5 skipped × 4 = 20

Lost by 5/10 — 0 wrong · 5 skipped

14. The equation $x^{(3/4)(\log_2^x)^2 + \log_2^x - (5/4)} = \sqrt{2}$ has

(A)at least one real solution (B)exactly four real solutions

(C)exactly two irrational solution (D)exactly four non-real solutions

Wrong: –

Skipped: M Charan Kumar Reddy, Ruperao Shritan, Kavuru Sahithi, M Jagadeesear Reddy, Aleti Dhanu Sai Sagar

Physics – high-impact questions

16 questions, worst first
cropped from the question paper

Q36 Kinematics – a(t) with modulus – area under a-t

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 9 wrong × 5 = 45

Lost by 9/10
9 wrong · 0 skipped

36. Starting from rest a particle moves in a straight line with acceleration $a = \{2 + |t-2|\}\, m/s^2$. Velocity of the particle at the end of 4s will be

(A) 16 m/s (B) 20 m/s (C) 8 m/s (D) 12 m/s

Wrong: Kolukula Hema Shiva Naga Hariharan, Ruperao Shritan, Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Nakka Mahidhar, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

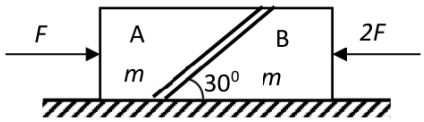
Skipped: –

Q43 Laws of Motion – Normal reaction across an inclined contact

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 7 wrong × 5 + 2 skipped × 4 = 43

Lost by 9/10
7 wrong · 2 skipped

43. Two blocks 'A" and 'B' each of mass 'm' are placed on a smooth horizontal surface. Two horizontal forces F and 2F are applied on both the blocks 'A' and 'B' respectively as shown in figure. The block A does not slide on block B. Then the normal reaction acting between the two blocks is



(A) F (B) $F/2$ (C) $F/\sqrt{3}$ (D) 3F

Wrong: Kolukula Hema Shiva Naga Hariharan, Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Skipped: Ruperao Shritan, Nakka Mahidhar

Q49 Kinematics – Free fall then deceleration – find height

Physics · Numerical · marking +4 / −1 · marks lost by the top 10: 2 wrong × 5 + 7 skipped × 4 = 38

Lost by 9/10
2 wrong · 7 skipped

49. A parachutist after bailing out falls 50m without friction. When parachute opens, it decelerates at $2m/\sec^2$. He reaches the ground with a speed of 3m/s. The height he bail out is nearly (in m) (Take g=10m/sec^2) .

Wrong: M Charan Kumar Reddy, Fareeha Shaik

Skipped: Ruperao Shritan, Kavuru Sahithi, M Jagadeesear Reddy, Nakka Mahidhar, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Q35 Kinematics – a = kv² – integrate to get v(s)

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 1 wrong × 5 + 8 skipped × 4 = 37

Lost by 9/10
1 wrong · 8 skipped

35. If a particle accelerates with $a = kv^2$ and initial velocity = u, then find velocity after displacement s

(A) $v = ue^{ks}$ (B) $v = u^{ks}e^{s}$ (C) $v = u^{k}e^{s}$ (D) $v = eu^{ks}$

Wrong: Nakka Mahidhar

Skipped: Kolukula Hema Shiva Naga Hariharan, Ruperao Shritan, Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Q47 Vectors – Net displacement from multi-leg journey

Physics · Numerical · marking +4 / −1 · marks lost by the top 10: 1 wrong × 5 + 8 skipped × 4 = 37

Lost by 9/10

1 wrong · 8 skipped

47. In going from one city to another , a car travels 75 km north, 60 km north –west and 20 km east. The magnitude(in km) of the vector drawn between the two cities is $\left(Take 1/\sqrt{2}=0.7\right)$

Wrong: Karthikeya Silver

Skipped: M Charan Kumar Reddy, Ruperao Shritan, Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Nakka Mahidhar, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Q41 Laws of Motion – Newton's 3rd law – fan blowing on own sail

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 7 wrong × 5 + 1 skipped × 4 = 39

Lost by 8/10

7 wrong · 1 skipped

41. On a stationary sail boat, air is blown at the sails from a fan attached to the boat. The boat will:

(A) Move forward

(B) move backward

(C) spin around

(D) remain stationary

Wrong: Kolukula Hema Shiva Naga Hariharan, Ruperao Shritan, Fareeha Shaik, M Jagadeesear Reddy, Nakka Mahidhar, Karthikeya Silver, Aleti Dhanu Sai Sagar

Skipped: Kavuru Sahithi

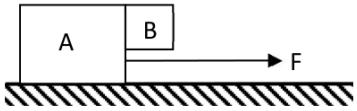
Q45 Laws of Motion – Contact force between pushed blocks (statement type)

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 6 wrong × 5 + 2 skipped × 4 = 38

Lost by 8/10

6 wrong · 2 skipped

45. Statement-1: Block A is moving on a horizontal surface towards right under action of force F. all surfaces are smooth. At the instant shown the force exerted by block A on block B is equal to net force on block B.



Statement-2: From Newton's third law, the force exerted by block A on B is equal in magnitude to force exerted by block B on A.

(A) Statement 1 is true, statement 2 is true, statement 2 is correct explanation for statement 1

(B) Statement 1 is true, statement 2 is true, statement 2 is not correct explanation for statement 1

(C) Statement 1 is true, statement 2 is false

(D) Statement 1 is false, statement 2 is true

Wrong: M Charan Kumar Reddy, Kavuru Sahithi, Fareeha Shaik, Nakka Mahidhar, Karthikeya Silver, Madipadiga Dheeraj

Skipped: Ruperao Shritan, Aleti Dhanu Sai Sagar

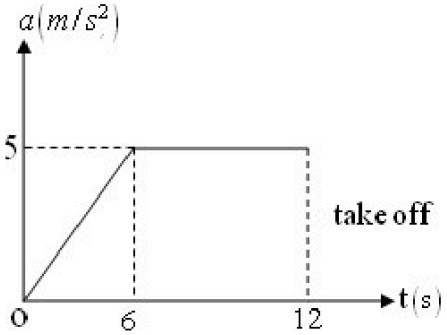
Q48 Kinematics – Distance from a-t graph

Physics · Numerical · marking +4 / −1 · marks lost by the top 10: 3 wrong × 5 + 5 skipped × 4 = 35

Lost by 8/10

3 wrong · 5 skipped

48. An experiment on the take off performance of an aeroplane shows that the acceleration varies as shown in the fig and it takes 12 sec to take off from a rest position.The distance along the runway covered by the aeroplane is(in m).



Wrong: Ruperao Shritan, Fareeha Shaik, M Jagadeesear Reddy

Skipped: M Charan Kumar Reddy, Kavuru Sahithi, Nakka Mahidhar, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Q38 Kinematics – Resistance ∝ v² – time inside plank

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 8 skipped × 4 = 32

Lost by 8/10

0 wrong · 8 skipped

38. Having gone through a plank of thickness h, a bullet changed its velocity from v_0 to v. Find the time of motion of the bullet in the plank assuming resistance force to be proportional to the square of the velocity

(A) $\left[\frac{h(v_0-v)}{vv_0 l\,n\frac{v_0}{v}}\right]$ (B) $\left[\frac{(v_0-v)}{hvv_0 l\,n\left(\frac{v_0}{v}\right)}\right]$ (C) $\left[\frac{vv_0 \ln\left(\frac{v_0}{v}\right)}{h(v_0-v)}\right]$ (D) $\left[\frac{\ln\frac{v_0}{v}}{vv_0(v_0-v)}\right]$

Wrong: –

Skipped: Kolukula Hema Shiva Naga Hariharan, M Charan Kumar Reddy, Ruperao Shritan, Kavuru Sahithi, M Jagadeesear Reddy, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Q42 Laws of Motion – Extra tension when bird leaves the wire

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 4 wrong × 5 + 3 skipped × 4 = 32

Lost by 7/10

4 wrong · 3 skipped

42. A bird dismounts on a stretched telegraph wire. The additional tension produced in the wire is

(A) Zero
(B) greater than the weight of the bird
(C) Less than the weight of the bird
(D) equal to the weight of the bird

Wrong: Kolukula Hema Shiva Naga Hariharan, M Charan Kumar Reddy, Kavuru Sahithi, M Jagadeesear Reddy

Skipped: Ruperao Shritan, Karthikeya Silver, Aleti Dhanu Sai Sagar

Q27 Vectors – Vector components under axis rotation

Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 2 wrong × 5 + 5 skipped × 4 = 30

Lost by 7/10

2 wrong · 5 skipped

27. The components of a vector along x and y directions are (n+1) and 1 respectively . If the co-ordinate system is rotated by an angle $\theta = 60^0$, then the components change to n and 3. The value of n is

(A) 2 (B) $\cos 60^0$ (C) $\sin 60^0$ (D) 3.5

Wrong: Kavuru Sahithi, Nakka Mahidhar

Skipped: Ruperao Shritan, Fareeha Shaik, M Jagadeesear Reddy, Karthikeya Silver, Aleti Dhanu Sai Sagar

Q30 **Laws of Motion** – Tension / normal reaction in two-block systems (matrix match)
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 5 wrong × 5 = 25

Lost by 5/10
5 wrong · 0 skipped

30. Column-I gives four different situations involving two blocks of mass m_1 and m_2 placed in different ways on a smooth horizontal surface as shown. In each of the situations horizontal forces F_1 and F_2 are applied on blocks of mass m_1 and m_2 respectively and also $m_2 F_1 < m_1 F_2$. Match the statements in column I with corresponding results in column-II

Column-I

(A) 

. Both the blocks are connected by massless inelastic string. The magnitude of tension in the string is

(B) 

. Both the blocks are connected by massless inelastic string. The magnitude of tension in the string is

(C) 

. The magnitude of normal reaction between the blocks is

(D) 

. The magnitude of normal reaction between the blocks is

Column-II

(p) $\frac{m_1 m_2}{m_1+m_2}\left(\frac{F_1}{m_1}-\frac{F_2}{m_2}\right)$

(q) $\frac{m_1 m_2}{m_1+m_2}\left(\frac{F_1}{m_1}+\frac{F_2}{m_2}\right)$

(r) $\frac{m_1 m_2}{m_1+m_2}\left(\frac{F_2}{m_2}-\frac{F_1}{m_1}\right)$

(s) $m_1 m_2\left(\frac{F_1+F_2}{m_1+m_2}\right)$

(A) (A) – r ; (B) – q ; (C) – q ; (D) – r
(B) (A) – q ; (B) – r ; (C) – r ; (D) – q
(C) (A) – q ; (B) – r ; (C) – q ; (D) – r
(D) (A) – q ; (B) – p ; (C) – r ; (D) – r

Wrong: Kolukula Hema Shiva Naga Hariharan, Kavuru Sahithi, Nakka Mahidhar, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj **Skipped:** –

Q40 **Laws of Motion** – Velocity over fixed distance vs mass ($v \propto 1/\sqrt{m}$)
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 5 wrong × 5 = 25

Lost by 5/10
5 wrong · 0 skipped

40. A constant force acts on a mass m at rest. Velocity acquired in travelling a fixed distance is directly proportional to

(A) $\sqrt{m}$ (B) m (C) $1/\sqrt{m}$ (D) $1/m$

Wrong: Fareeha Shaik, M Jagadeesear Reddy, Nakka Mahidhar, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj **Skipped:** –

Q28 **Vectors** – Resultant ⊥ smaller vector; find magnitudes
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 4 wrong × 5 + 1 skipped × 4 = 24

Lost by 5/10
4 wrong · 1 skipped

28. The simple sum of two co-initial vectors is 16 units . Their vector sum is 8 units. The resultant of the vectors is perpendicular to the smaller vector . The magnitudes of the two vectors are

(A) 22units and 14 units
(B) 4 units and 12 units
(C) 6 units and 10 units
(D) 8 units and 12 units

Wrong: Kavuru Sahithi, Fareeha Shaik, Nakka Mahidhar, Madipadiga Dheeraj **Skipped:** M Jagadeesear Reddy

Q29 **Vectors – Angles between three vectors summing to zero**
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 4 wrong × 5 + 1 skipped × 4 = 24

Lost by 5/10
4 wrong 1 skipped

29. Given: $\vec{a}+\vec{b}+\vec{c}=0$. Out of the three vectors $\vec{a},\vec{b}$ and $\vec{c}$, two are equal in magnitude. The magnitude of the third vector is $\sqrt{2}$ times that of either of the two having equal magnitude. The angles between the vector are

(A) $90^0,135^0,135^0$ (B) $30^0,60^0,90^0$

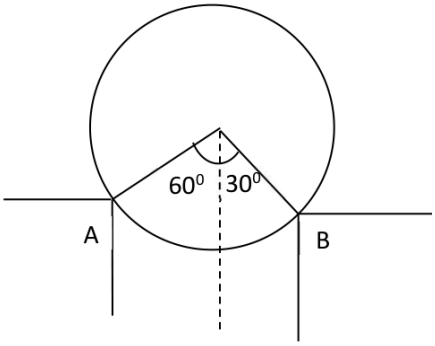
(C) $45^0,45^0,90^0$ (D) $45^0,60^0,90^0$

Wrong: Kolukula Hema Shiva Naga Hariharan, M Jagadeesear Reddy, Nakka Mahidhar, Madipadiga Dheeraj **Skipped:** Aleti Dhanu Sai Sagar

Q44 **Laws of Motion – Normal reactions on a cylinder resting on edges**
Physics · Single correct · marking +4 / −1 · marks lost by the top 10: 4 wrong × 5 + 1 skipped × 4 = 24

Lost by 5/10
4 wrong 1 skipped

44. A cylinder of mass M and radius R is resting on two corner edges A and B as shown in the figure. The normal reaction at the edges A and B are (neglect friction)



(A) $N_A=\sqrt{2}N_B$ (B) $N_B=\sqrt{3}N_A$ (C) $N_A=\sqrt{3}N_B$ (D) $N_B=\frac{2\sqrt{3}Mg}{5}$

Wrong: Ruperao Shritan, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj **Skipped:** Nakka Mahidhar

Chemistry – high-impact questions

6 questions, worst first
cropped from the question paper

Q67 Chemical Bonding – Strongest ionic bond (charge and size)

Chemistry · Single correct · marking +4 / −1 · marks lost by the top 10: 10 wrong × 5 = 50

Lost by 10/10
10 wrong · 0 skipped

67. Which combination will give strongest ionic bond?

(A) K^+ and Cl^- (B) K^+ and O^{2-}

(C) Ca^{2+} and Cl^- (D) Ca^{2+} and O^{2-}

Wrong: Kolukula Hema Shiva Naga Hariharan, M Charan Kumar Reddy, Ruperao Shritan, Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Nakka Mahidhar, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Skipped: –

Q75 Chemical Bonding – Hybrid orbitals + lone pairs in NH_3

Chemistry · Numerical · marking +4 / −1 · marks lost by the top 10: 4 wrong × 5 + 3 skipped × 4 = 32

Lost by 7/10
4 wrong · 3 skipped

75. According to VBT, consider the molecule NH_3. The central nitrogen atom undergoes hybridisation before bond formation.
If the number of hybrid orbitals formed is x, and the number of lone pairs present on nitrogen is y, find:
x + y

Wrong: Kolukula Hema Shiva Naga Hariharan, Ruperao Shritan, Nakka Mahidhar, Karthikeya Silver

Skipped: M Charan Kumar Reddy, M Jagadeesear Reddy, Madipadiga Dheeraj

Q65 Periodic Classification – Largest radius among isoelectronic ions

Chemistry · Single correct · marking +4 / −1 · marks lost by the top 10: 5 wrong × 5 + 1 skipped × 4 = 29

Lost by 6/10
5 wrong · 1 skipped

65. Which of the following has largest radius ?

(A) Mg^{2+} (B) Na^+ (C) O^{2-} (D) F^-

Wrong: Kolukula Hema Shiva Naga Hariharan, M Charan Kumar Reddy, Fareeha Shaik, Nakka Mahidhar, Karthikeya Silver

Skipped: M Jagadeesear Reddy

Q55 Mole Concept & Basics – Laws of chemical combination

Chemistry · Single correct · marking +4 / −1 · marks lost by the top 10: 4 wrong × 5 + 1 skipped × 4 = 24

Lost by 5/10
4 wrong · 1 skipped

55. 12 g carbon combines with 64 g sulphur to form CS_2. 12 g carbon also combines with 32 g oxygen to form CO_2. 10 g sulphur combines with 10 g oxygen to form SO_2. These data illustrate the -

(A) Law of multiple proportions
(B) Law of definite proportions
(C) Law of reciprocal proportions
(D) Law of gaseous volumes.

Wrong: Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Karthikeya Silver

Skipped: Ruperao Shritan

Q59 **Chemical Bonding** – VBT, orbital overlap, hybridisation statements
Chemistry · Single correct · marking +4 / −1 · marks lost by the top 10: 4 wrong × 5 + 1 skipped × 4 = 24

Lost by 5/10
4 wrong · 1 skipped

59. Statement I: According to VBT, a covalent bond is formed by the overlap of half-filled atomic orbitals containing electrons with opposite spins.
Statement II: Greater the extent of effective orbital overlap, stronger is the covalent bond formed.
Statement III: Hybridisation involves the mixing of atomic orbitals of comparable energy to form equivalent hybrid orbitals.
Which of the following is correct?
(A) Only Statement I is correct
(B) Statements I and II are correct
(C) Statements I, II and III are correct
(D) Statements II and III are correct

Wrong: Kavuru Sahithi, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj **Skipped:** M Jagadeesear Reddy

Q53 **Mole Concept & Basics** – Valence electrons in N^{3-} (mole → electrons)
Chemistry · Single correct · marking +4 / −1 · marks lost by the top 10: 3 wrong × 5 + 2 skipped × 4 = 23

Lost by 5/10
3 wrong · 2 skipped

53. If N_A is Avogadro number, then the number of valence electrons in 4.2 g of N^{3-} ions is
(A) 2.4 N_A (B) 4.2 N_A (C) 1.6 N_A (D) 3.2 N_A

Wrong: Kavuru Sahithi, Karthikeya Silver, Madipadiga Dheeraj **Skipped:** Fareeha Shaik, Nakka Mahidhar